

Student Course Management System Report

Student & Course Information

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- **Course Name:** Foundations of Software Development
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- **Github Link:** <https://github.com/maiar38/C-Capstone>

1. Objective

The objective of this project is to develop a C++ console application that manages students and their enrolled courses. The system allows users to perform operations such as adding, removing, searching, and displaying students, as well as managing course enrollments.

2. Learning Outcomes

- Object-Oriented Programming (OOP)
- Use of STL containers
- Implementation of searching and sorting algorithms
- File handling in C++
- Writing modular and maintainable code

3. Program Description

The system is a menu-driven console application that enables users to add, remove, search, and display students, enroll them in courses, show courses, sort by GPA, and save/load data from a file.

4. System Design (OOP Concepts)

Classes used:

Person:

- Attribute: name
- Methods: constructor, getName()

Student (inherits from Person):

- Attributes: id, gpa, courses (set<string>)
- Methods: getId(), getGpa(), enrollCourse(), showCourses(), saveToFile()

StudentManager:

- Attribute: vector<Student> students
- Methods: addStudent(), removeStudent(), findStudent(), displayAll(), enrollCourse(), showCourses(), sortByGPA(), saveToFile(), loadFromFile(), searchByName()

5. STL Usage

Containers:

- vector (to store students)
- set (to store courses and avoid duplicates)

Algorithms:

- sort() (to sort students by GPA)
- remove_if() (to remove students)

6. Functional Requirements Implementation

All required features are implemented:

- Add student
- Remove student
- Search by ID
- Display all students
- Enroll in course
- Show courses
- Sort by GPA

7. Constraints Handling

- Unique ID ensured
- GPA validated between 0.0 and 4.0
- Invalid input handled
- Duplicate courses avoided using set

8. File Handling (Bonus Feature)

The system supports saving and loading data from a file (students.txt), including student ID, name, GPA, and courses.

9. Additional Features

- Search by name
- Automatic saving
- Menu-based interface
- Input validation

10. Challenges Faced

- Handling file format
- Managing pointers
- Avoiding duplicate courses

11. Conclusion

This project demonstrates OOP principles, STL usage, and file handling. The system is efficient and meets all requirements.